

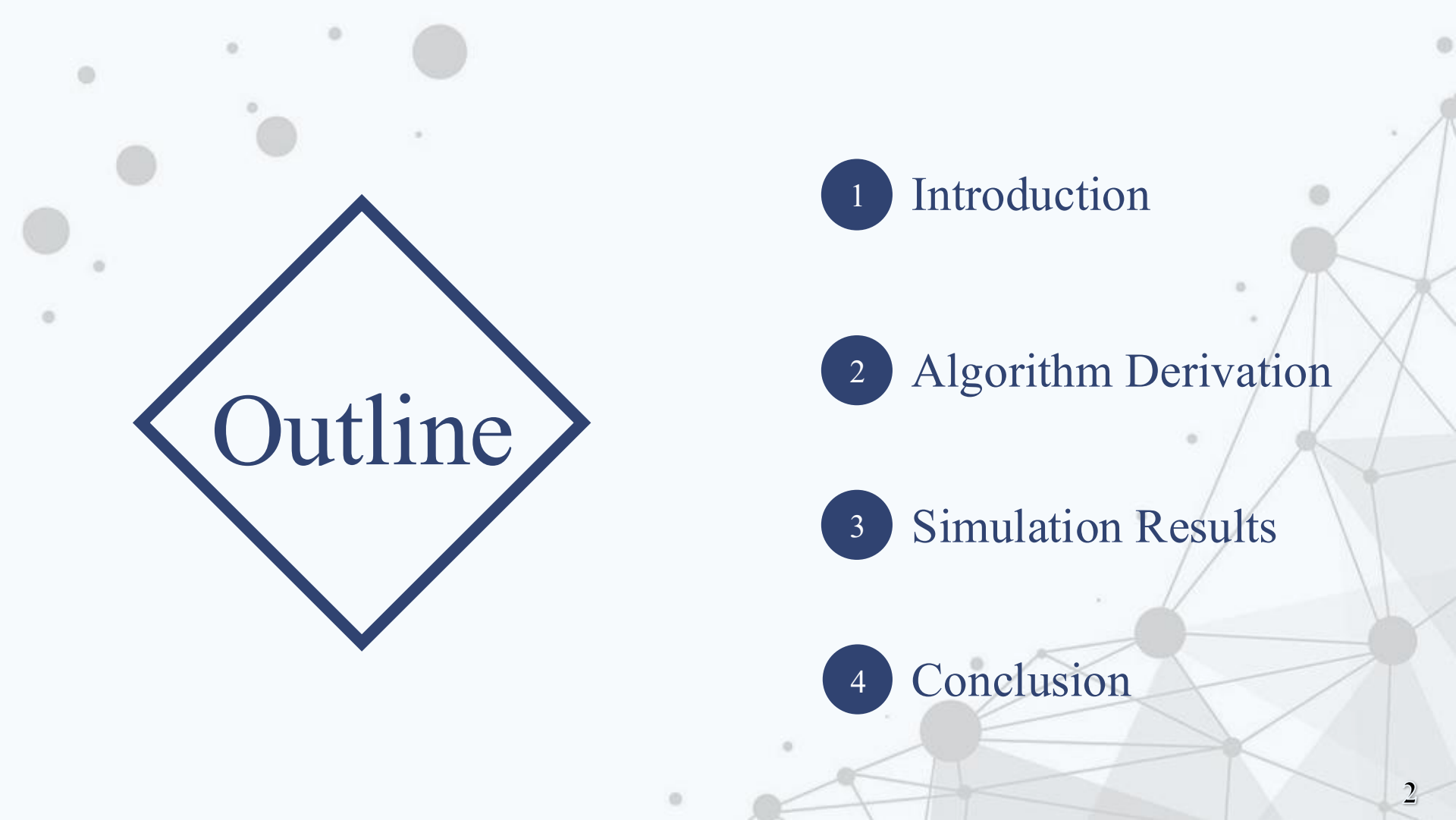
A Bayesian Step Least Mean Squares Algorithm for Gaussian Signals

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Date: 2026/05/29

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Outline

- 1 Introduction
- 2 Algorithm Derivation
- 3 Simulation Results
- 4 Conclusion



Introduction

Some results show that the LMS is equivalent to the Kalman filter (KF) if the state noise of the KF and the step of the LMS algorithm are carefully selected.

Introduction

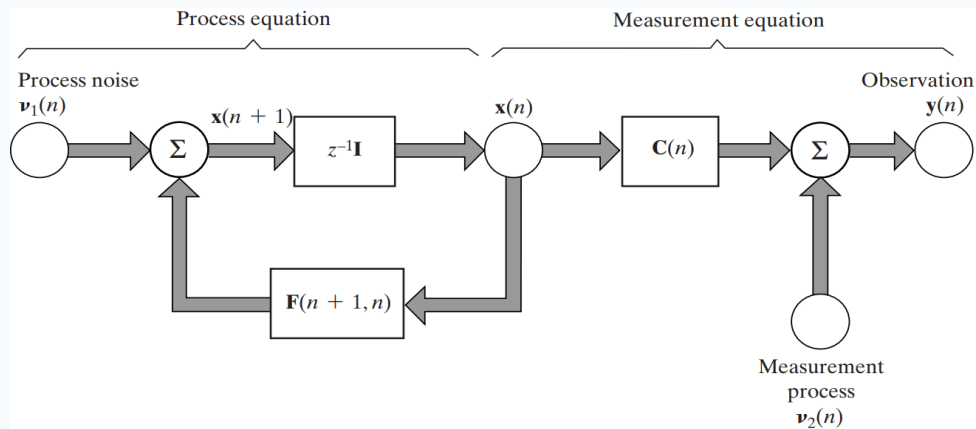
- Selecting the step size of the LMS algorithm is a classical problem.
- LMS with a **time-varying step** size can be related to the Kalman filter.
- The optimal step size depends on:
 - coefficient estimation error variance q_w
 - measurement noise variance q_v

Introduction

- Existing methods such as NLMS, EM, and JO-LMS can adaptively choose the step size.
- [27] Many methods assume white noise or known noise statistics.
- This paper proposes a Bayesian approach with fewer prior assumptions.

Introduction

- LMS adaptive filtering can be interpreted using the Kalman state-space model.



$$x(n+1) = F(n+1, n)x(n) + v_1(n) \quad (\text{process equation})$$

$$y(n) = C(n)x(n) + v_2(n) \quad (\text{measurement equation})$$



$$R(n) = C(n)K(n+1, n)C^H(n) + Q_2(n)$$



Algorithm Derivation

The method used in this paper is to use the joint probability density function to find q_w, q_V and use the Gaussian approximation method to get the update equation.

Step Size Calculation[5]

LMS algorithm: $\mathbf{w}(n) = \mathbf{w}(n-1) + \mu e(n)\mathbf{u}(n)$

$$\begin{aligned}e(n) &= d(n) - \mathbf{u}^T(n)\mathbf{w}(n-1) \\ \mathbf{u}(n) &= [u(n) \dots u(n-N+1)]^T \\ \mathbf{w}(n) &= [w_0(n) \dots w_{N-1}(n)]^T\end{aligned}$$

Step size from: $\mu(n) = \frac{1}{p_t(n) + q_v(n)/q_w(n)}$

$$\begin{aligned}p_t(n) &= \mathbf{u}^T(n)\mathbf{u}(n) \\ q_v &: \text{measurement noise variance} \\ q_w &: \text{variance of the current estimate} \\ &\quad \text{of each coefficient}\end{aligned}$$

Optimum coefficient: $\mathbf{w}(n) = E[\mathbf{x}(n) | q_w(n), q_v(n), \mathbf{w}(n-1)]$ $\mathbf{x}(n) = \mathbf{w}_0(n)$: optimal values

↳ for a carefully chosen state noise

Step Size Calculation

- $q_w(n)$ and $q_v(n)$ are unknown

➤ $\mathbf{w}_b(n) = \mathbb{E}[\mathbf{x}(n) | \mathbf{w}(n-1)]$

$$= \int \int^{\mathbf{E}} [\mathbf{x}(n) | q_w(n), q_v(n), \mathbf{w}(n-1)]$$

$$\rho_q(q_w(n), q_v(n), n) dq_w(n) dq_v(n)$$

$\rho_q(q_w(n), q_v(n), n)$: the joint PDF
of $q_w(n)$ and $q_v(n)$

Law of total probability:

$$P(A) = \int P(A|B) \cdot P(B) dB$$

➤ $\mathbf{w}_b(n) = \int \int^{\mathbf{w}} (n) \rho_q(q_w(n), q_v(n), n) dq_w(n) dq_v(n)$

➤ $\mathbf{w}_b(n) = \mathbf{w}_b(n-1) + \mu_b(n) e(n) \mathbf{u}(n)$

Since only $\mu(n)$ depends
on $q_w(n)$ and $q_v(n)$

$$\mu_b(n) = \mathbb{E}[\mu(n)] = \int \int \mu(n) \rho_q(q_w(n), q_v(n), n) dq_w(n) dq_v(n)$$

State Variables PDF

$$\begin{aligned}\mathbf{q}(n) &= [q_w(n), q_v(n)]^T \\ \mathbf{a}(n) &= [p_t(n), \mathbf{1}]^T \\ E[e^2(n)] &= \mathbf{a}^T(n)\mathbf{q}(n) \\ e(n) &: \text{zero-mean Gaussian}\end{aligned}$$

- Calculate $q_w \rho_q(q_w(n), q_v(n), n) \Rightarrow$ Posterior PDF is updated using Bayes rule.

$$\begin{aligned}\rho_{q(n)|n}(\mathbf{q}) &= \rho_{q(n)|e(n), e(n-1), \dots}(\mathbf{q}) \\ &= \frac{\rho_{e(n)|q(n)}(\mathbf{q}) \rho_{q(n)|n-1}(\mathbf{q})}{\rho_{e(n)|n-1}(\mathbf{q})} = k \frac{e^{-(e^2(n)/2\mathbf{a}^T(n)\mathbf{q})}}{\sqrt{2\pi \mathbf{a}^T(n)\mathbf{q}}} \rho_{q(n)|n-1}(\mathbf{q})\end{aligned}$$

Bayes' rule:

$$f_{X|Y}(x|y) = \frac{f_X(x) \cdot f_{Y|X}(y|x)}{f_Y(y)}$$

$$\rho_{q|n}(\mathbf{q}) = k \left(\prod_{i=1}^n \frac{e^{-(e^2(i)/2\mathbf{a}^T(i)\mathbf{q})}}{((2\pi)^2 \mathbf{q}^T \mathbf{a}(i) \mathbf{a}^T(i) \mathbf{q})^{1/4}} \right) \rho_{q|0}(\mathbf{q})$$

Assume

$$\rho_{q(n)|n-1}(\mathbf{q}) = \rho_{q(n)|n-1}(\mathbf{q})$$

$$\rho_{q(n)|n}(\mathbf{q}) \simeq \hat{\rho}_q(\mathbf{q}, \mathbf{D}(n), \mathbf{c}(n), n) = k_0 \frac{e^{-(n/2\mathbf{c}^T(n)\mathbf{q})}}{((2\pi)^2 \mathbf{q}^T \mathbf{D}(n) \mathbf{q})^{n/4}}$$

$$\mathbf{D}(n) = \begin{bmatrix} d_{1,1}(n) & d_{1,2}(n) \\ d_{2,1}(n) & d_{2,2}(n) \end{bmatrix}$$

$$\mathbf{c}(n) = [c_1(n), c_2(n)]^T$$

State Variables PDF

- Calculate $q_w \rho_q(q_w(n), q_v(n), n) \Rightarrow$ Gaussian approximation is used to simplify PDF

Γ = Gamma function

Change of random variable:

$$\rho_y(y) = \int \rho_x(x) \left| \frac{dy}{dx} \right| \delta(y - g(x)) dx$$

Approximated by a Gaussian function

$$\text{mean: } z_1 = z_0 - \frac{\delta_1(z_0)}{\delta_2(z_0)}$$

$$\text{variance: } q_z = -\frac{1}{\delta_2(z_0)}$$

$$\text{let } p_z = \frac{1}{q_z}$$

$\text{erf}(x)$ = Gauss error function

$$\rho_\mu(z) = \int \frac{\hat{\rho}_q(q, D, c, n)}{z^2 / q_w} dq_w$$

$$= \frac{1}{4} \pi^{-n/2} \Gamma\left(\frac{n}{2} - 2\right) \left(\frac{-c_2 z p_t + c_1 z + c_2}{n} \right)^{(n-4)/2} \\ (d_{2,2}(z p_t - 1)^2 + z(z d_{1,1} - 2 d_{1,2}(z p_t - 1)))^{-n/4}$$

$$E[\mu] \simeq \int_0^{1/p_t} z \hat{\rho}_\mu(z) dz$$

$$a = \frac{\sqrt{p_z}(1 - z_1 p_t)}{\sqrt{2} p_t}, \quad b = \frac{z_1 \sqrt{p_z}}{\sqrt{2}},$$

$$E[\mu] \simeq z_1 + \frac{-2e^{-a^2} + 2e^{-b^2}}{\sqrt{2\pi p_z}(\text{erf}(a) + \text{erf}(b))}$$

Measure Update Equations

- Determine how to update $\mathbf{D}$ and $\mathbf{c}$

$\mathbf{A}^{-1}$: multivariate Gaussian covariance matrix
mean: $\mathbf{q}_0 = \mathbf{A}^{-1}\mathbf{b}$

$$\triangleright \hat{\rho}_a(\mathbf{q}, \mathbf{D}, \mathbf{c}, n) \simeq \tilde{\rho}_a(\mathbf{q}, \mathbf{A}, \mathbf{b}) = k_1 e^{\mathbf{b}^T \mathbf{q} - \frac{1}{2} \mathbf{q}^T \mathbf{A} \mathbf{q}}$$



Approximated by a Gaussian function

- (i) The values of $\mathbf{q}$ that maximizes both functions are equal.

$$\mathbf{q}_0 = \frac{\mathbf{D}^{-1}\mathbf{c}}{\mathbf{c}^T \mathbf{D}^{-1} \mathbf{c}} = \mathbf{A}^{-1}\mathbf{b}$$

- (ii) The Hessian matrices of the logarithm of both functions ($\ln \hat{\rho}_q$ and $\ln \tilde{\rho}_q$) are equal at the maximum point $\mathbf{q}_0$.

$$\frac{n}{2} \mathbf{D}(\mathbf{c}^T \mathbf{D}^{-1} \mathbf{c}) = \mathbf{A}$$

Measure Update Equations

- Determine how to update $\mathbf{D}$ and $\mathbf{c}$

➤ $\mathbf{D} = \mathbf{A}, \xi = \mathbf{b}^T \mathbf{A}^{-1} \mathbf{b}$

$$\mathbf{c} = \frac{\mathbf{b}}{\xi}, n = 2\xi.$$

$$\begin{aligned}\xi &= \mathbf{b}^T \mathbf{q}_0 = \mathbf{q}_0^T \mathbf{A} \mathbf{q}_0 \\ \zeta &= \mathbf{a}^T \mathbf{q}_0\end{aligned}$$

➤ $\rho_{\mathbf{q}(n)|n}(\mathbf{q}) \simeq k \frac{e^{-\left((e^2(n) - \mathbf{a}^T(n) \mathbf{q}(n))^2 / (4\zeta^2(n))\right)}}{\sqrt{4\pi \zeta^2(n)}} \tilde{\rho}_{\mathbf{q}}(\mathbf{q})$

mean: $\mathbf{a}^T(n) \mathbf{q}(n)$
variance: $2\zeta^2$

➤ $\mathbf{A}(n+1) = \mathbf{A}(n) + \frac{\mathbf{a}(n) \mathbf{a}^T(n)}{2\zeta^2(n)}$

$\mathbf{b}(n+1) = \mathbf{b}(n) + \frac{\mathbf{a}(n) e^2(n)}{2\zeta^2(n)}.$



the measure update equations

Time Update Equations

Table 1 Proposed algorithm

Initialisation

$$\mathbf{w}(0) = [1, 0, \dots, 0]^T \quad (31)$$

$$\mathbf{q}_o(0) = [1, 1]^T \quad (32)$$

$$\mathbf{A} = 0, \mathbf{b} = 0 \quad (33)$$

$$\delta = \text{small number} \quad (34)$$

Iterate

$$\mathbf{e}(n) = \mathbf{d}(n) - \mathbf{u}(n)^T \mathbf{w}(n) \quad (35)$$

$$p_t(n) = \mathbf{u}(n)^T \mathbf{u}(n), \quad p(n) = p_t(n)/N \quad (36)$$

$$\mathbf{a}(n) = [p_t(n), 1]^T \quad (37)$$

$$\boldsymbol{\zeta}(n) = \mathbf{a}^T(n) \mathbf{q}_o(n) \quad (38)$$

$$\mathbf{A}(n) = \mathbf{A}(n-1) + \frac{\mathbf{a}(n) \mathbf{a}^T(n)}{2\boldsymbol{\zeta}^2(n)} \quad (39)$$

$$\mathbf{b}(n) = \mathbf{b}(n-1) + \frac{\mathbf{a}(n) e^2(n)}{2\boldsymbol{\zeta}^2(n)} \quad (40)$$

$$\mathbf{q}_o(n) = (\mathbf{A}(n) + \delta \mathbf{I})^{-1} \mathbf{b}(n) \quad (41)$$

$$\boldsymbol{\xi}(n) = \mathbf{b}(n)^T \mathbf{q}_o(n) \quad (42)$$

$$\mathbf{c}(n) = \mathbf{b}(n)/\boldsymbol{\xi}(n) \quad (43)$$

$$n_x(n) = 2\boldsymbol{\xi}(n) \quad (44)$$

$$[q_w(n), q_v(n)]^T = \mathbf{q}_o(n) \quad (45)$$

$$z(n) = \min \left(\max \left(\frac{q_w(n)}{q_w(n)p_t(n) + q_v(n)}, 0 \right), \frac{1}{p_t} \right) \quad (46)$$

$$\mu(n) = \begin{cases} z(n) & n_x \leq 4 \\ 1/(2p_t) & n_x > 4 \text{ and } \delta_2(n) > 0 \\ \text{calculate (17)} & \text{otherwise} \end{cases} \quad (47)$$

$$\mathbf{w}(n+1) = \mathbf{w}(n) + \mu(n) \mathbf{u}(n) e(n) \quad (48)$$

$$\gamma_1(n) = 1 \quad \text{if } q_w(n) > 0 \quad \text{else} \quad -1 \quad (49)$$

$$\gamma_2(n) = 1 \quad \text{if } q_v(n) > 0 \quad \text{else} \quad -1 \quad (50)$$

$$\mathbf{C}(n) = \begin{bmatrix} \gamma_1(n)(1 - \alpha\mu(n)p(n))^2 & \alpha\gamma_2(n)\mu^2(n)p(n) \\ 0 & \gamma_2(n) \end{bmatrix} \quad (51)$$

$$\hat{\mathbf{A}}(n+1) = (\mathbf{C}(n)^{-1})^T \mathbf{A}(n) \mathbf{C}(n)^{-1} \quad (52)$$

$$\hat{\mathbf{b}}(n+1) = (\mathbf{C}(n)^{-1})^T \mathbf{b}(n) \quad (53)$$

$$k_t(n) = 1 \quad \text{if } \text{trace}(\hat{\mathbf{A}}(n+1)) \leq T \quad \text{else} \quad \frac{T}{\text{trace}(\hat{\mathbf{A}}(n+1))} \quad (54)$$

$$\mathbf{A}(n+1) = k_t \hat{\mathbf{A}}(n+1) \quad (55)$$

$$\mathbf{b}(n+1) = k_t \hat{\mathbf{b}}(n+1) \quad (56)$$

Time Update Equations

Table 2 Proposed Algorithm subroutine to calculate (17)

$$\delta_a = n(-2a_{1,2}(zp_t - 1) + z(a_{1,1} - 2a_{1,2}p_t)) \quad (63)$$

$$+ 2a_{2,2}p_t(zp_t - 1) + za_{1,1})$$

$$\delta_b = 4(a_{2,2}(zp_t - 1)^2 + z(za_{1,1} - 2a_{1,2}(zp_t - 1)))$$

$$\delta_1(z) = \frac{(n-4)(c_1 - c_2p_t)}{2(-c_2zp_t + c_1z + c_2)} - \frac{\delta_a}{\delta_b}$$

$$\delta_c = n(-2a_{1,2}(zp_t - 1) + z(a_{1,1} - 2a_{1,2}p_t)) \quad (64)$$

$$+ 2a_{2,2}p_t(zp_t - 1) + za_{1,1})^2$$

$$\delta_d = 4(a_{2,2}(zp_t - 1)^2 + z(za_{1,1} - 2a_{1,2}(zp_t - 1)))^2$$

$$\delta_2(z) = -\frac{n(2a_{2,2}p_t^2 - 4a_{1,2}p_t + 2a_{1,1})}{4(a_{2,2}(zp_t - 1)^2 + z(za_{1,1} - 2a_{1,2}(zp_t - 1)))}$$

$$+ \frac{\delta_c}{\delta_d} - \frac{(n-4)(c_1 - c_2p_t)^2}{2(-c_2zp_t + c_1z + c_2)^2}$$

$$z_1 = z - \delta_1(z)/\delta_2(z) \quad (65)$$

$$p_z = -\delta_2(z) \quad (66)$$

$$a = \frac{\sqrt{p_z}(1 - z_1p_t)}{\sqrt{2}p_t}, \quad b = \frac{z_1\sqrt{p_z}}{\sqrt{2}}, \quad (67)$$

$$q_w(n)$$

$$\text{if } a + b < \delta \quad (68)$$

$$0$$

$$\text{elseif } |a| > 10 \wedge |b| > 10 \wedge \text{sign}(a) \langle 0 \wedge \text{sign}(b) \rangle 0$$

$$z_1 + \frac{2 - 2e^{-b^2 + a^2}}{\sqrt{2\pi p_z} \left(\frac{e^{-1/(2a^2)}}{\sqrt{\pi a}} + \frac{e^{-b^2 + a^2 - 1/(2b^2)}}{\sqrt{\pi b}} \right)}$$

$$\text{else if } |a| > 10 \wedge |b| > 10 \wedge \text{sign}(a) > 0 \wedge \text{sign}(b) < 0 \quad (69)$$

$$z_1 - \frac{2 - 2e^{-a^2 + b^2}}{\sqrt{2\pi p_z} \left(\frac{e^{-1/(2b^2)}}{\sqrt{\pi b}} + \frac{e^{-a^2 + b^2 - 1/(2a^2)}}{\sqrt{\pi a}} \right)}$$

$$\text{elseif } |a| > 3 \wedge |b| > 3 \wedge \text{sign}(a)\text{sign}(b) < 0 \quad (70)$$

$$z_1 + \frac{2e^{-a^2} - 2e^{-b^2}}{\sqrt{2\pi p_z} \{ \text{sign}(a)\text{erfc}(|a|) + \text{sign}(b)\text{erfc}(|b|) \}}$$

$$\text{else}$$

$$z_1 + \frac{-2e^{-a^2} + 2e^{-b^2}}{\sqrt{2\pi p_z} \{ \text{erf}(a) + \text{erf}(b) \}} \quad (71)$$



Simulation Results

This low value will make the $C(n)$ matrix close to identity and the $A(n)$ matrix close to singular, resulting in there being insufficient information to determine q_w and q_v accurately in a short time.

Simulation Results

Estimation of q_v and q_w . Average of 50 simulations

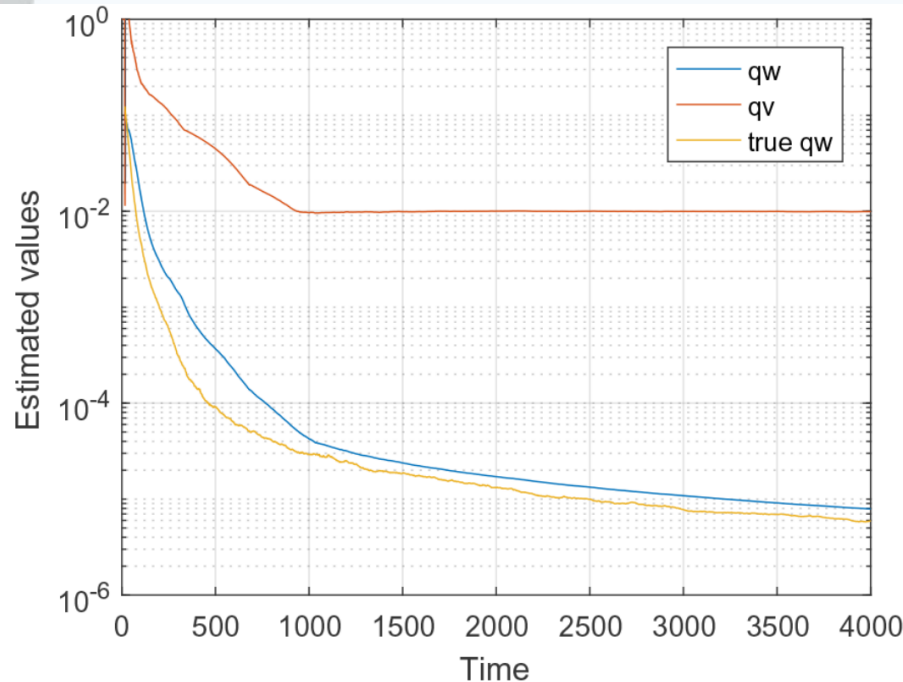


Table 3 General simulation parameters

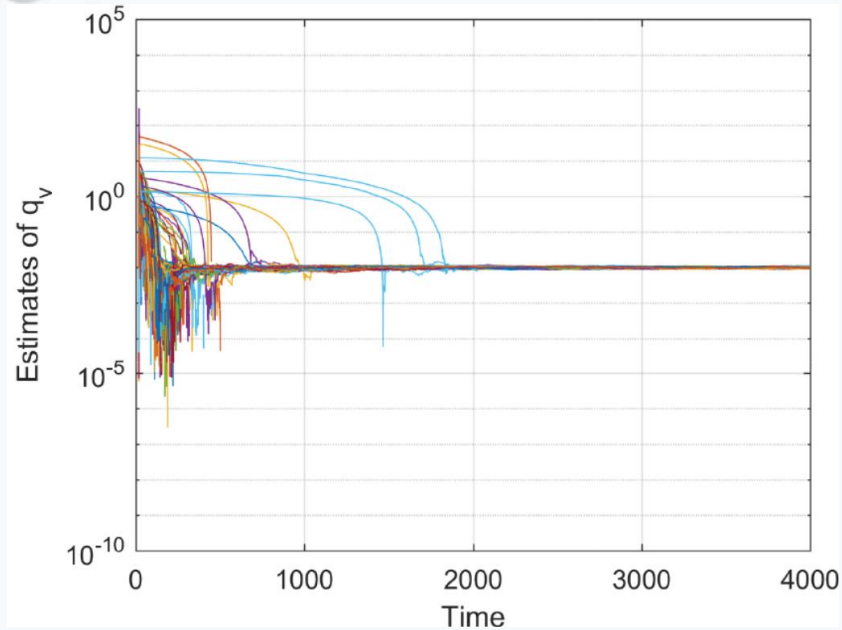
M	100	number of Monte Carlo runs
N	16	filter size
q_v	0.01	measurement noise variance
q_u	1.00	reference signal variance
s_w	1.00	coefficient standard deviation
h	[1, 0.5]	reference filter impulse response

Table 4 General algorithm parameters

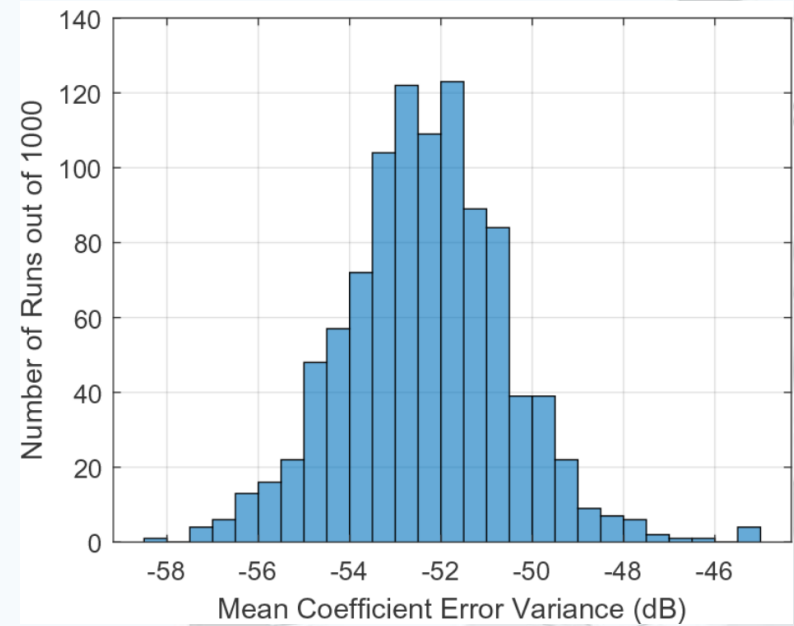
α	0.33	eigenvalue spread compensation factor
T	10^{11}	maximum trace of A

Simulation Results

Estimates of q_v . Results of 100 single Monte Carlo runs.

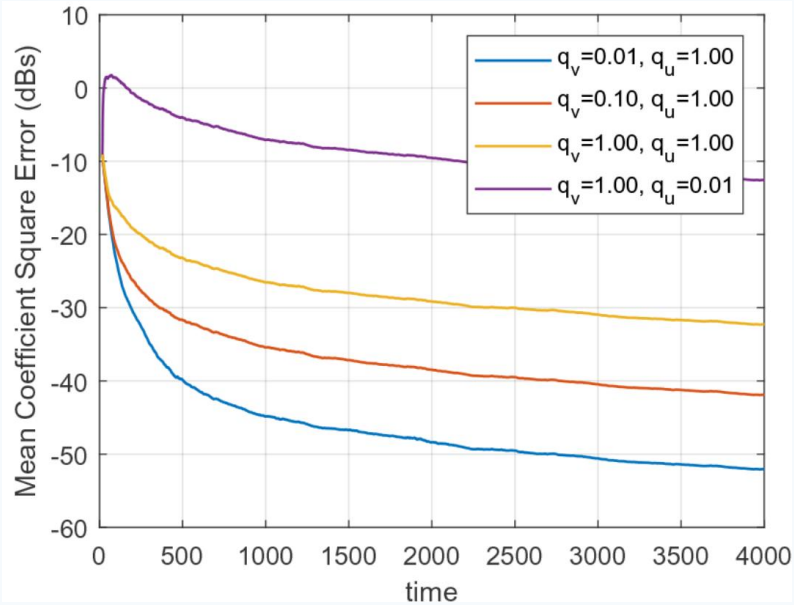


Histogram of the final value for the mean coefficient error variance

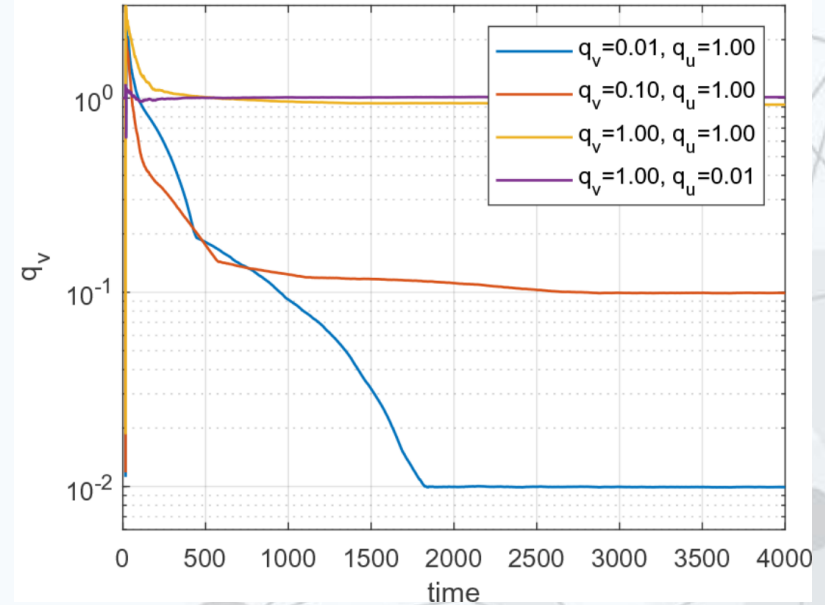


Simulation Results

Mean coefficient estimation error for different values of true- q_v and q_u .

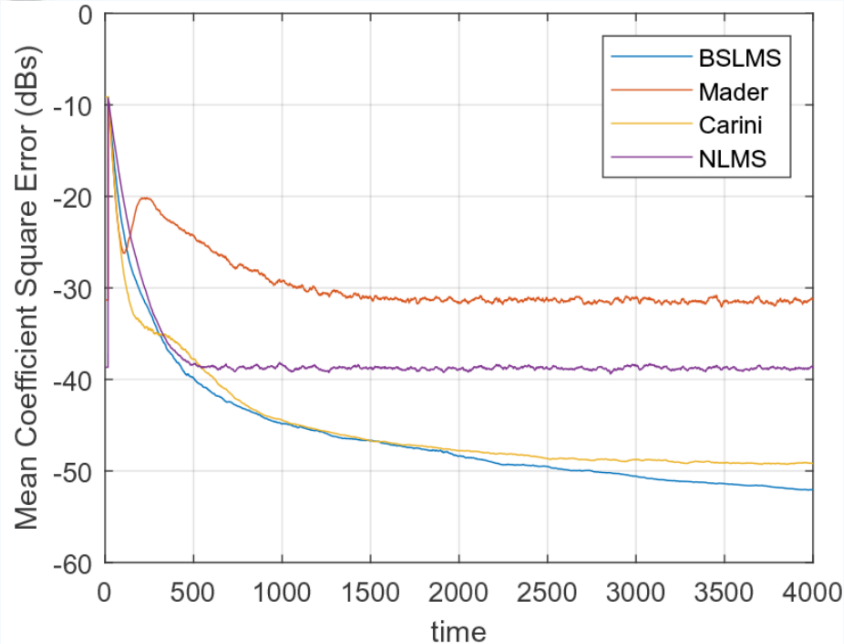


Estimate of the measurement noise variance q_v for different values of true- q_v and q_u .

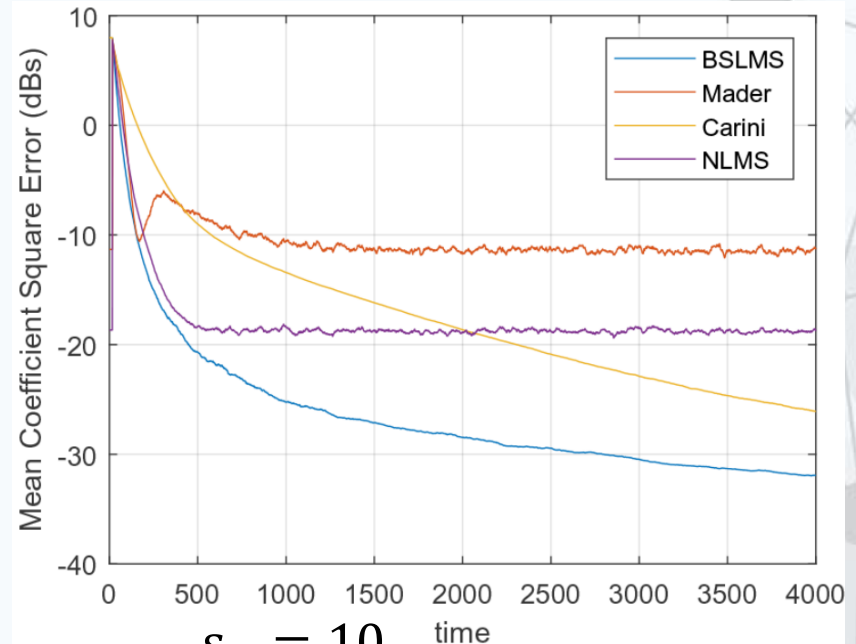


Simulation Results

Comparison with other algorithms with
optimized parameters



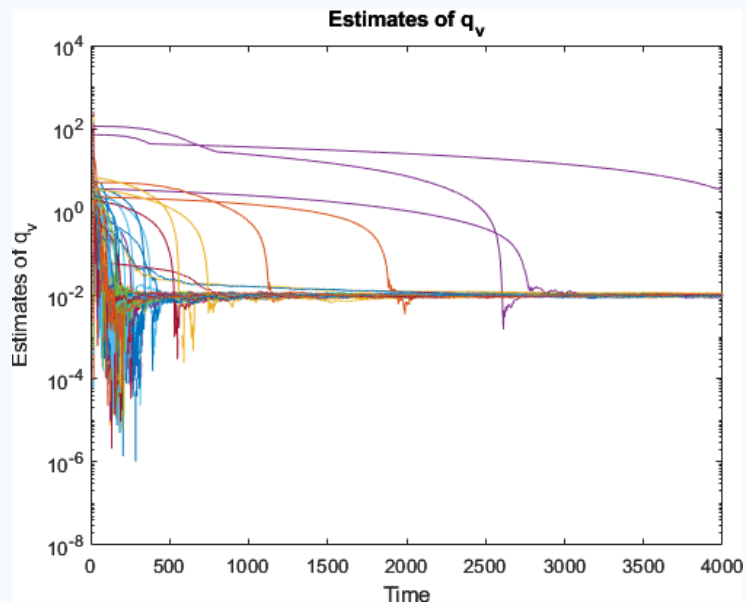
Comparison with other algorithms with
non-optimized parameters



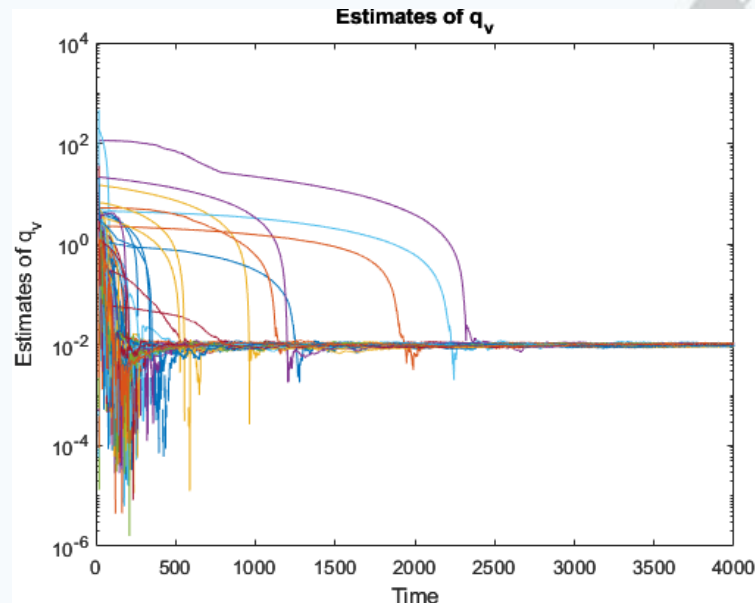
$$s_w = 10$$
$$q_v = 1$$

Change Initial Value

Simulation results in paper

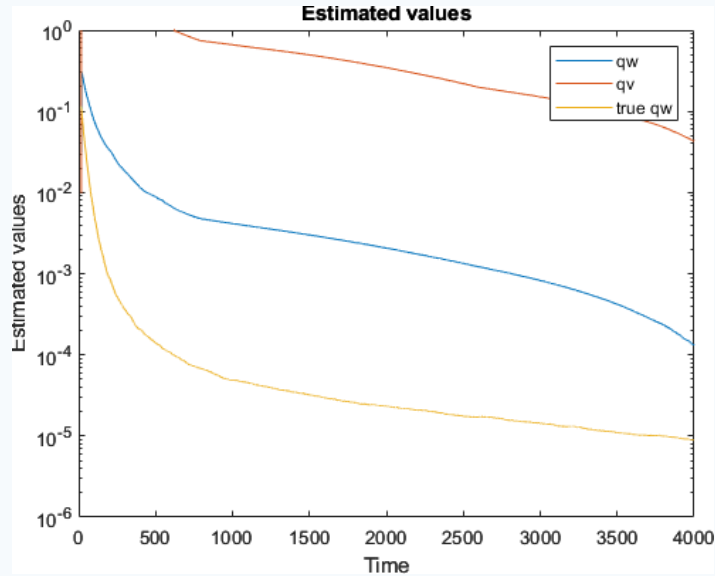


Our simulation results

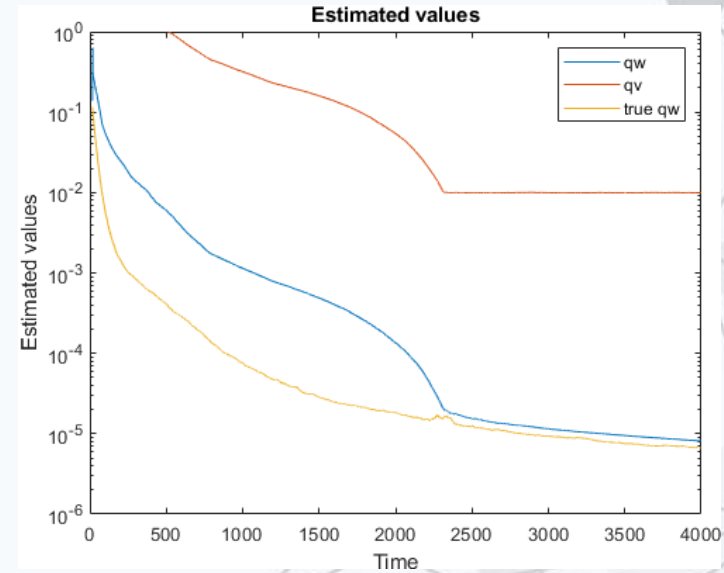


Change Initial Value

Simulation results in paper



Our simulation results



$$s_w = 10$$
$$q_v = 1$$



Conclusion

The step size
is selected using the expected value of the step of the KLMS
algorithm by integrating the corresponding PDF.

Conclusion

- A Bayesian method is proposed for adaptive LMS step-size selection.
- It estimates unknown variances online.
- Faster convergence and lower steady-state error are achieved.
- Robust under different initial values and noise levels.

The background features a light gray abstract design. On the left, there are several circles of varying sizes, some solid and some outlined. On the right, a network of thin gray lines connects various circular nodes, creating a web-like structure. The overall aesthetic is clean and modern.

End of My Presentation

Thank you for listening